Nicolas Anguita

+44 7305738716| na658@cam.ac.uk | linkedIn/nicolas-anguita | Google Scholar Profile

EDUCATION

University of Cambridge Cambridge, UK | Jan 2025 - July 2028

PhD in Machine Learning and Theoretical Neuroscience

- Computational and Biological Learning Lab, Department of Engineering. Supervised by **Yashar Ahmadian** and **Andrew Saxe** (Gatsby Computational Neuroscience Unit), advised by **Guillaume Hennequin**.
- Studying the theoretical principles of continual and representation learning, aiming to understand how the brain retains and reuses knowledge across tasks and applying this to novel AI systems. **Interests:** Learning Dynamics, Probabilistic ML, Reinforcement Learning, LLMs.
- **Rafael del Pino Excellence Fellow** (≈19 recipients annually), awarded by H.M. the King of Spain.
- Selected as **one of 30 fully funded participants (from 2,400+ applicants)** for the MBZUAI Winter School in GenAI and Representation Learning.
- Thinking About Thinking Fellow. Supervisor, 3F8 Inference (3rd-year Engineering). Invited talks: Oxford Random Matrix Theory group, Gatsby Computational Neuroscience Unit, **AI Safety Group**.

Imperial College London London, UK | Sep 2020 - Jul 2024

MEng Mathematics and Computer Science - First Class (>80% average, 92% in Master's thesis)

- Distinguished Project award. **Minimax Labs prize awarded for an excellent project showcasing algorithmic development and engineering, or scientific computation.**
- Courses include: Probability Theory, Stochastic Differential Equations, Machine Learning, Philosophy of Mind.
- Master Thesis: Learning Dynamics of Linear Neural Networks, joint with Imperial College and **Gatsby Computational Neuroscience Unit**.

City of London School London, UK | Sep 2018 - Jul 2020

6A* A-Level/equivalent

- Combination of UK (4 A-Levels) and US (7 APs, self-study). **Met university requirements by the age of 15.**
- British Mathematical and Chemistry Olympiad. AP Scholar with Distinction.

PUBLICATIONS

- **N Anguita**, Francesco Locatello, Andrew M. Saxe et al. ***A Theory of How Pretraining Shapes Inductive Bias in Fine-Tuning***. Published in **ICML 2026**. Also presented at **Cosyne 2026**, where it received a competitive grant for top ranked papers from the *Simons Foundation*.
- **N Anguita***, CCJ Domine* et al. ***From Lazy to Rich: Exact Learning Dynamics in Deep Linear Networks***. Published in **ICLR 2025**. Also published in two NeurIPS 2024 workshops: M3L (Mathematics for Machine Learning) and UniReps (Unifying Representations in Neural Models).
- **N Anguita**, ..., Andrew M. Saxe, Yashar Ahmadian. ***Implicit Bias Theory of Machine Unlearning***: Exactly solvable theory of **tamper-resistant unlearning** with direct practical recommendations for robust unlearning, relevant to **AI safety**. Submitted to **UniReps 2026** workshop on Unifying Representations in Neural Models. Large-model experiments in progress for a main-conference submission.
- Connal Garrod, **Nicolas Anguita**, ..., Christos Thrampoulidis: ***Exact Solutions and Saddle-to-Saddle Dynamics in Nonlinear Matrix Factorization***: Extends exact learning dynamics beyond linear networks; reveals synchronised mode emergence and transient disruption of learned concepts. Accepted to **NeurIPS 2026** workshop NeurReps

ONGOING RESEARCH PROJECTS

- **Affordances and World Models**: A theory of how the actions available to an agent (its affordances) shape the internal representations (world models) it forms, using Transformers and RNNs trained on next-token prediction as an experimental and theoretical testbed.

SELECTED WORK EXPERIENCE

XANTIUM GROUP - TUDOR CORPORATION | Quant Research Intern London, UK | Jul 2026 - Sep 2026

- Short-Term Macro team. Applying machine learning methods from my PhD research to real-world time-series domains: highly noisy, non-stationary financial signals. Industry exposure that has sharpened my research agenda toward practical, deployable theory. Technologies: PyTorch, JAX, scikit-learn, internal libraries.

BREVAN HOWARD | Trading Intern London, UK | Jul 2023 – Aug 2023

- Developed and evaluated machine-learning trading models using Random Forest and NLP methods; explored volatility-regime detection via Hidden Markov Models.